Introduction to Inverse Problems and Imaging - 02624

Week 1, September 4, 2024

Main topics

- Introduction to the course
- Inverse problems and imaging - examples and model problems
- Well-posedness and Ill-posedness
- Discretization of integral equations

Material

Section 1.1-1.2 in [Kirsch] and section 3.1 in [PCH]. Both books are uploaded to learn. One is from findit, one is from the author. Please do not distribute.

Tentative program (non-standard schedule)

13:00-14:00 Introduction to the course: inverse and ill-posed problems

14:00-15:45 Exercise session

15:45-17:00 Discretization of integral equations

The goal is to establish a common vocabulary and get started on inverse problems.

Exercises

You are highly encouraged to discuss and solve the exercises in groups. We will return to exercises 6-8 next week.

1. (a) Given a function x in $C([0,1])$, find a solution y in $C^1([0,1])$ to the equation $y'(t) = x(t)$. Equip the equation with boundary conditions $y(0) = 0$. Is there a solution and is it unique?

   (b) Write y as an integral operator acting on x. What is the forward, and what is the inverse problem?

   (c) Show that the forward problem is well-posed for $X = Y = C([0,1])$, and for $X = C([0,1]), Y = C^1([0,1])$. Is the inverse problem also well-posed for both scenarios?

2. Consider in the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1.001 \\ 0 & 0 \end{bmatrix}.$$

Consider the vectors

$$b_1 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \quad b_3 = \begin{bmatrix} 2 \\ 2.001 \\ 0 \end{bmatrix}.$$

For what j's is the problem

$$Ax = b_j$$

well-posed?

3. In the Hilbert space $L^2(0, \pi)$ the functions $\varphi_n(t) = \sqrt{\frac{2}{\pi}} \sin(nt), n = 1, 2, \ldots$, constitute an orthonormal basis. Let's define the operator $K_N \in B(L^2(0, \pi))$ ($N > 1$ is fixed) acting on $f(t) = \sum_{n=1}^{\infty} c_n \varphi_n(t)$ by

$$K_N f(t) = \sum_{n=1}^{N} \frac{c_n}{n} \varphi_n(t).$$

Let $g \in L^2(0, \pi)$ be our data/measurement and consider the inverse problem of solving $Kf = g$. Is the problem wellposed?
Hint: You might want to consider $g = \varphi_M$ for $M > N$.

4. In the spirit of the previous exercise, take $N = \infty$ and define

$$Kf(t) = \sum_{n=1}^{\infty} \frac{c_n}{n} \varphi_n(t).$$

Show that $K \in B(L^2(0, \pi)$; consequently the forward problem is well-posed. Is the inverse problem of solving $Kf = g$ for some given $g \in L^2(0, \pi)$ wellposed?
Hint: Look at the stability for the inverse by considering $g = \varphi_M$ as $M \to \infty$. Can you establish a stability bound $\|f\|_{L^2(0,\pi)} \leq C\|g\|_{L^2(0,\pi)}$ for some $C > 0$ independent of f, g?

5. Consider the operator K on $L^2(-1, 1)$ defined by

$$Kx(t) = \frac{1}{2t} \int_{-t}^{t} x(s) \mathrm{d}s, \quad 0 < t < 1,$$

and the inverse problem

$$Kx(t) = y(t)$$

   (a) Discuss what information y carries about x?

   (b) Show that $z(t) = ty(t)$ satisfies

$$z' = x_{\text{even}}, \quad z(0) = 0,$$

   where x_{even} is the even part of x.

(c) Is the inverse problem $Kx = y$ well-posed for, say, assuming x is an even function and $ty \in L^2(0,1)$ or $ty \in C^1([0,1])$?

6. Consider the forward heat equation taking initial condition $f \in L^2(0,\pi)$ to terminal condition $g \in L^2(0,\pi)$. Show that the problem has a unique solution. Establish using Fourier series and Parseval's identity the stability estimate $\|g\|_{L^2(0,\pi)} \leq \|f\|_{L^2(0,\pi)}$.

7. Consider the backwards heat equation defined for functions g in the range of the forward operator. Is the solution unique? Can you establish a stability bound $\|f\|_{L^2(0,\pi)} \leq C\|g\|_{L^2(0,\pi)}$ for some $C > 0$ independent of f, g?

8. Let B be the unit ball in the plane and suppose that $f \in C(\mathbb{R}^2)$ is a radially symmetric function with support inside B (i.e. $f(x,y) = 0$ for $x^2 + y^2 \geq 1$.) Show that the Radon transform

$$R(\rho,\varphi) = \int_{l(\rho,\varphi)} f(x,y)\,\mathrm{d}s$$

is independent of φ.